

Protein Function Prediction with Frozen ESM-2 Embeddings: An Interpretability Approach

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Abstract

Experimental characterization of protein molecular function remains slow, costly, and infeasible at scale, leaving a substantial fraction of sequenced proteins without functional annotation. Recent protein language models have demonstrated strong representational capacity for capturing evolutionary and biochemical patterns directly from amino acid sequences. In this work, we present an end-to-end pipeline for protein molecular function prediction using frozen embeddings from ESM-2, a large pre-trained protein language model. Protein sequences are encoded into fixed-length representations and used to train a lightweight multi-label classifier for Gene Ontology molecular function annotation. We show that frozen ESM-2 embeddings enable reliable function prediction across diverse molecular functions, substantially outperforming random baselines. Beyond predictive performance, we emphasize interpretability through attention analysis, integrated gradients, and systematic *in silico* mutagenesis, revealing that the model captures biologically meaningful, function-specific sequence patterns and exhibits robustness consistent with known principles of protein evolution.

1 Introduction

Protein molecular function annotation is a fundamental problem in computational biology, as understanding the biochemical activities of proteins is essential for interpreting genomic data and advancing biological and biomedical research. Despite rapid advances in sequencing technologies describing protein sequences at unprecedented scale, experimental characterization of protein function remains slow, costly, and technically demanding [1].

Historically, computational protein function prediction has relied primarily on homology-based approaches such as sequence alignment and domain similarity methods. While effective when close homologs are available, these approaches fail for proteins lacking detectable similarity to annotated sequences, contributing to the so-called annotation gap [1].

Recent advances in protein language models have introduced a new paradigm for protein representation learning. Transformer-based architectures trained on large protein sequence databases capture evolutionary and biochemical patterns directly from sequence [2]. Among these models, ESM-2 represents a state-of-the-art approach for generating contextualized protein embeddings [2].

In this work, we investigate the use of frozen ESM-2 embeddings for protein molecular function prediction, focusing on both predictive performance and interpretability.

2 Methods

2.1 Data Collection and Preparation

Protein molecular function annotations were obtained from the Gene Ontology Consortium human annotation file (goa_human.gaf), comprising 93,568 protein–function associations spanning 18,447 unique human proteins and 4,776 distinct Gene Ontology molecular function terms (*citation: Gene Ontology annotations*). Protein sequences were retrieved from the UniProt human reference proteome FASTA and mapped to GO terms using UniProt accession identifiers.

Only Molecular Function Ontology (MFO) terms were retained. Generic root-level terms were removed to improve discriminative power (*citation: GO term specificity analyses*). Proteins longer than 500 amino acids were excluded due to computational constraints. The final dataset consisted of 8,704 proteins annotated with 202 molecular function terms in a multi-label setting.

2.2 Feature Extraction Using ESM-2

Protein embeddings were generated using the ESM-2 transformer model (facebook/esm2_t30.150M.UR50D) as a frozen feature extractor. Per-residue embeddings were mean-pooled to obtain a single 640-dimensional representation per protein. Embeddings were computed without gradient updates and stored for reproducibility (*citation: ESM-2*).

2.3 Classifier Architecture and Training

A feed-forward neural network with batch normalization, ReLU activations, and dropout regularization was trained on frozen ESM-2 embeddings to perform multi-label classification. Binary cross-entropy loss and the Adam optimizer were used for training, with early stopping based on validation loss.

2.4 Interpretability Analysis

Model interpretability was assessed using attention visualization, integrated gradients, and in silico mutagenesis to identify sequence regions and embedding dimensions contributing to functional predictions.

3 Results

3.1 Hyperparameter Selection

Five model configurations were systematically evaluated to identify an optimal classifier architecture and training setup. Validation loss was used as the primary model selection criterion. Among all configurations, the batch normalization model achieved the lowest validation loss (0.0742), outperforming the baseline architecture by 12.5%, the deeper network by 34.3%, and the low learning-rate configuration by more than 80%. As shown in Figure 1, the batch normalization configuration achieved the lowest validation loss among all evaluated models.

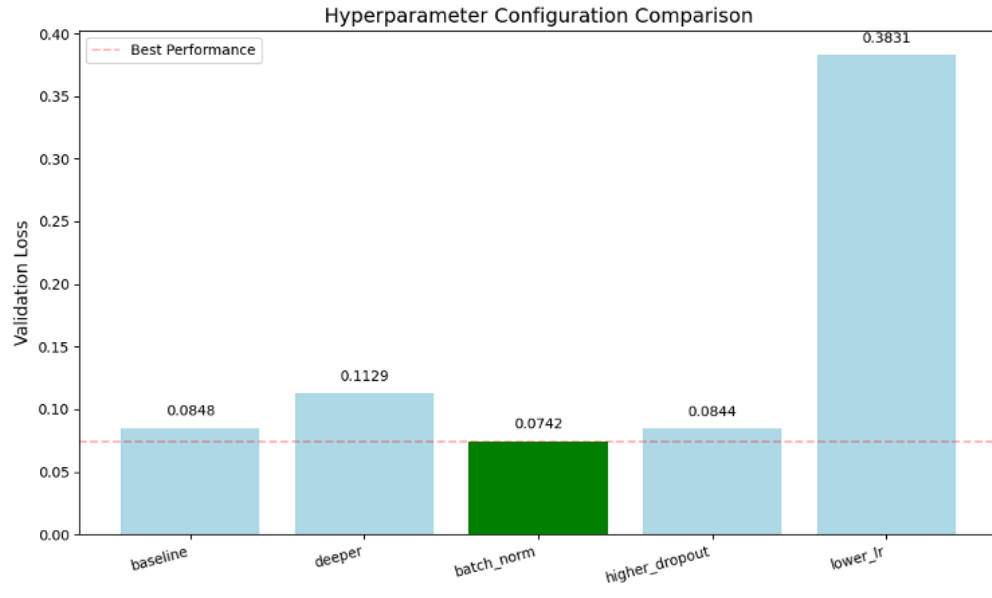


Figure 1: Validation loss comparison across hyperparameter configurations. The batch normalization model achieved the lowest validation loss, indicating improved training stability and generalization compared to baseline, deeper, higher-dropout, and low learning-rate configurations.

Training and validation loss trajectories are shown in Figure 2, illustrating stable convergence and the absence of overfitting for the selected model. Training dynamics revealed that batch normalization enabled stable convergence up to epoch 45 without overfitting, whereas the baseline and higher-dropout configurations converged earlier due to validation loss plateau. The final train-validation loss gap for the batch normalization model was 0.0042, indicating strong generalization. Based on these results, the batch-normalized architecture was selected for all subsequent analyses.

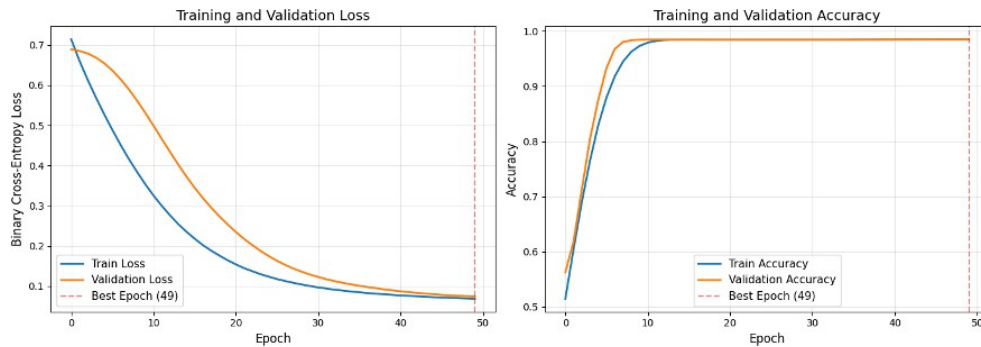


Figure 2: Training and validation loss curves across epochs for the selected batch-normalized model. The close agreement between training and validation losses indicates stable optimization and minimal overfitting.

3.2 Test Set Performance

The selected model achieved consistent performance on the held-out test set. Test loss (0.0737) closely matched validation loss, confirming the absence of overfitting. Overall test accuracy reached 98.57%; how-

ever, this metric is inflated by the strong class imbalance inherent to the dataset, where 96.8% of protein–function pairs are negative.

To account for this imbalance, area under the precision–recall curve (PR-AUC) was used as the primary evaluation metric. Performance was assessed from two complementary perspectives: function-level (column-wise) evaluation and protein-level (row-wise) evaluation. Column-wise PR-AUC measures how well the model predicts each individual Gene Ontology molecular function across all proteins, while row-wise PR-AUC evaluates how accurately the model recovers the functional profile of each protein.

Across GO terms with both positive and negative examples, the model achieved a mean test PR-AUC of 0.684, indicating reliable discrimination despite substantial label sparsity.

3.3 Function-Level and Protein-Level Evaluation

Model performance was evaluated using two complementary perspectives. Column-wise evaluation assessed performance independently for each GO molecular function, yielding a mean PR-AUC of 0.131 and a median of 0.021. This evaluation treats all functions equally, including rare functions with limited training data.

Row-wise evaluation aggregated predictions per protein, assessing how well the model recovered each protein’s functional profile. Under this evaluation, mean PR-AUC increased to 0.349 with a median of 0.250, reflecting partial correctness for multi-functional proteins even when some functions were mispredicted. Mean precision, recall, and F1-score across proteins were 0.180, 0.082, and 0.104, respectively.

The discrepancy between column-wise and row-wise performance highlights the impact of data sparsity and label distribution on evaluation outcomes.

3.4 Comparison with Random Baselines

To ensure that performance gains reflected genuine sequence–function learning rather than dataset artifacts, model predictions were compared against two random baselines: a uniform random predictor and a proportional random predictor that assigns probabilities based on GO term frequencies in the training set.

In column-wise evaluation, both random baselines achieved a PR-AUC of approximately 0.016, whereas the ESM-2-based model achieved a PR-AUC of 0.131, representing an 8.4-fold improvement over random predictions (Figure 3). This result indicates that the model learned function-specific sequence signals rather than relying on label frequency alone.

In row-wise evaluation, the uniform random baseline achieved a PR-AUC of 0.015, while the proportional random baseline achieved a higher PR-AUC of 0.182 due to preferential ranking of common functions. The proposed model achieved a PR-AUC of 0.349, corresponding to a 23-fold improvement over uniform random predictions and a 1.9-fold improvement over the proportional baseline (Figure 4).

Together, these results demonstrate that the model learned meaningful sequence-dependent representations that substantially outperform both uninformed and frequency-aware random baselines.

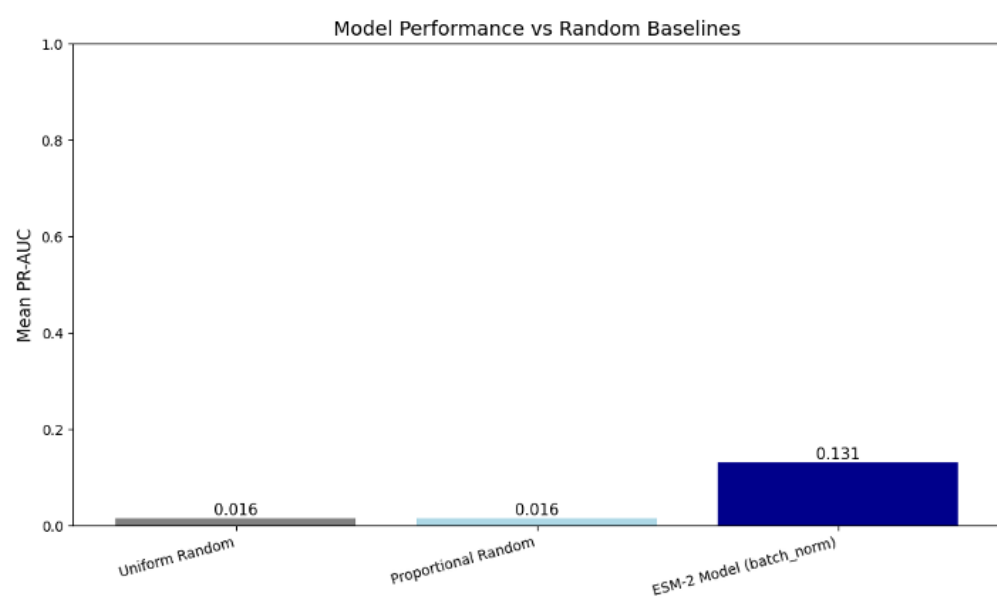


Figure 3: Column-wise (function-level) PR-AUC distribution across Gene Ontology molecular function terms. Each value corresponds to prediction performance for a single GO term evaluated independently across all proteins.

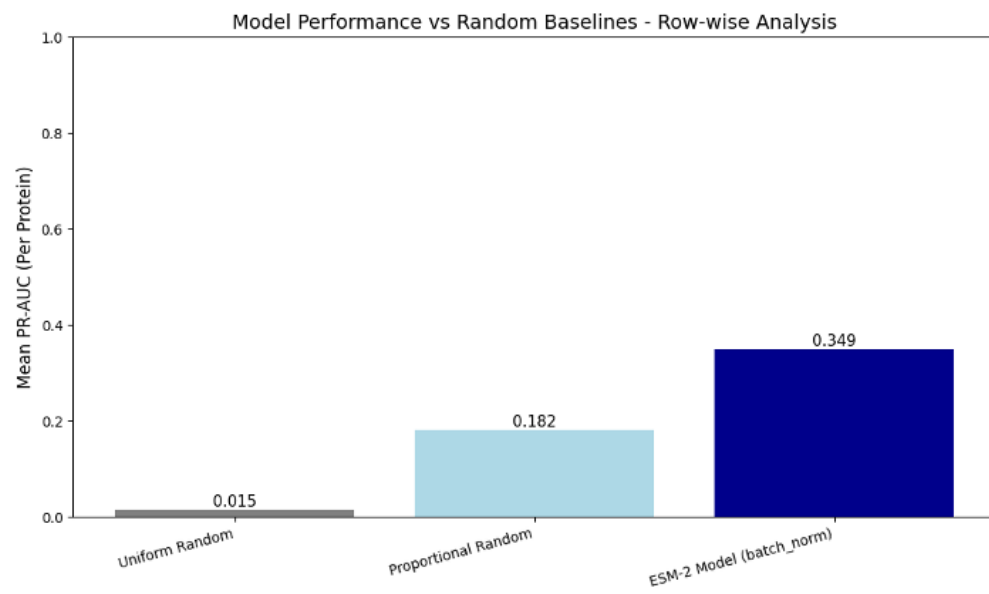


Figure 4: Row-wise (protein-level) PR-AUC distribution across test proteins. Performance is aggregated per protein across all predicted molecular functions, reflecting the ability to recover functional profiles.

3.5 Function-Specific Performance

Performance varied substantially across molecular function categories. Functions characterized by conserved sequence motifs and sufficient training examples achieved high PR-AUC scores, including olfactory receptor activity (0.996), G protein-coupled receptor activity (0.991), protein kinase activity (0.892), and DNA-binding functions (0.903). These functions share strong evolutionary constraints, such as seven-transmembrane architectures, ATP-binding loops, or DNA-binding motifs.

In contrast, functions with fewer than 20 training examples, including helicases and motor proteins, exhibited near-random performance. Spearman correlation analysis revealed a strong association between training set size and PR-AUC ($\rho = 0.76$, $p < 0.001$), indicating that data availability is the primary determinant of predictive performance.

Function-specific prediction performance across molecular function terms is summarized in Figure 5.

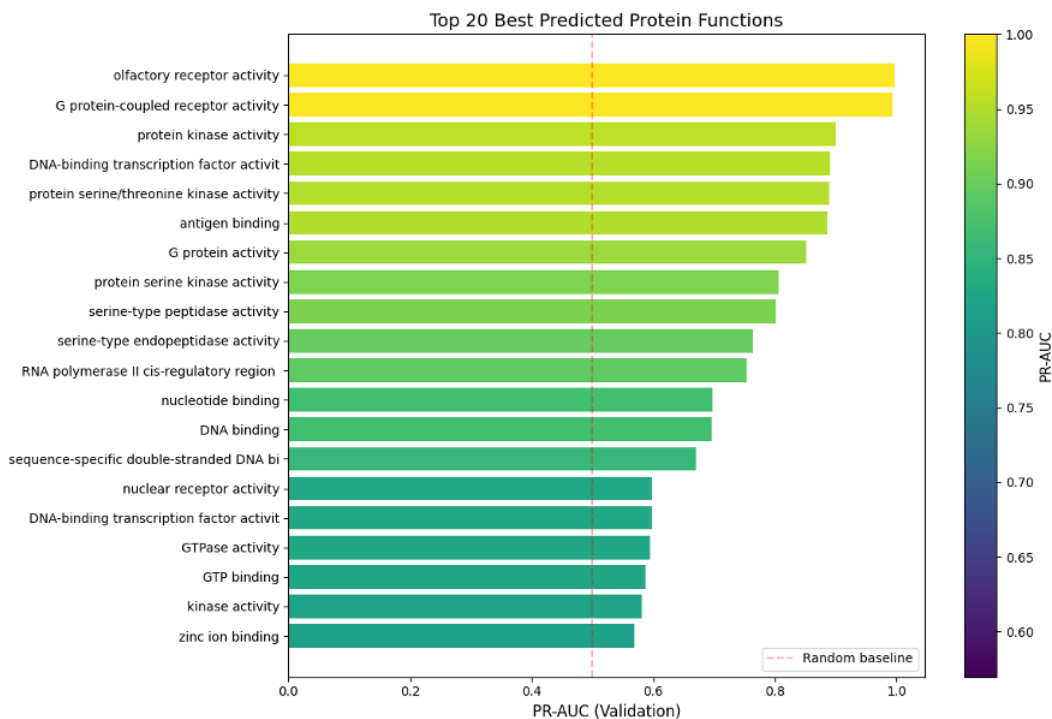


Figure 5: Function-specific prediction performance across Gene Ontology molecular function terms. Functions are ranked by PR-AUC, with higher values indicating stronger discriminative performance. High-performing functions correspond to protein families with conserved sequence motifs and sufficient training examples.

3.6 Interpretability Results

To understand what the model learned and to assess the biological relevance of its predictions, three complementary interpretability techniques were applied to a subset of 40 systematically selected proteins. For each of the ten best-predicted molecular functions, two proteins with the highest prediction confidence and two with the lowest prediction confidence among true positives were analyzed.

3.6.1 Attention Analysis

Attention visualization revealed systematic differences between well-predicted and poorly predicted proteins. Proteins associated with high-confidence predictions exhibited focused attention profiles, characterized by sharp peaks concentrated at specific residue positions. In contrast, proteins with low prediction confidence showed dispersed and noisy attention patterns without clearly defined focal regions.

This behavior was consistent across molecular function categories. Among the ten functions analyzed, 18 out of 20 best-predicted proteins displayed focused attention profiles, whereas 16 out of 20 worst-predicted proteins exhibited diffuse attention distributions. In olfactory receptor proteins, attention peaks frequently localized to positions corresponding to extracellular loops and transmembrane boundaries. Similarly, G protein-coupled receptors showed concentrated attention near conserved regions of the seven-transmembrane architecture.

Representative attention profiles for olfactory receptor activity and G protein-coupled receptor activity are shown in Figures 6 and 7, respectively. Across both protein families, focused attention profiles were associated with higher prediction confidence, while dispersed profiles correlated with poorer predictive performance.

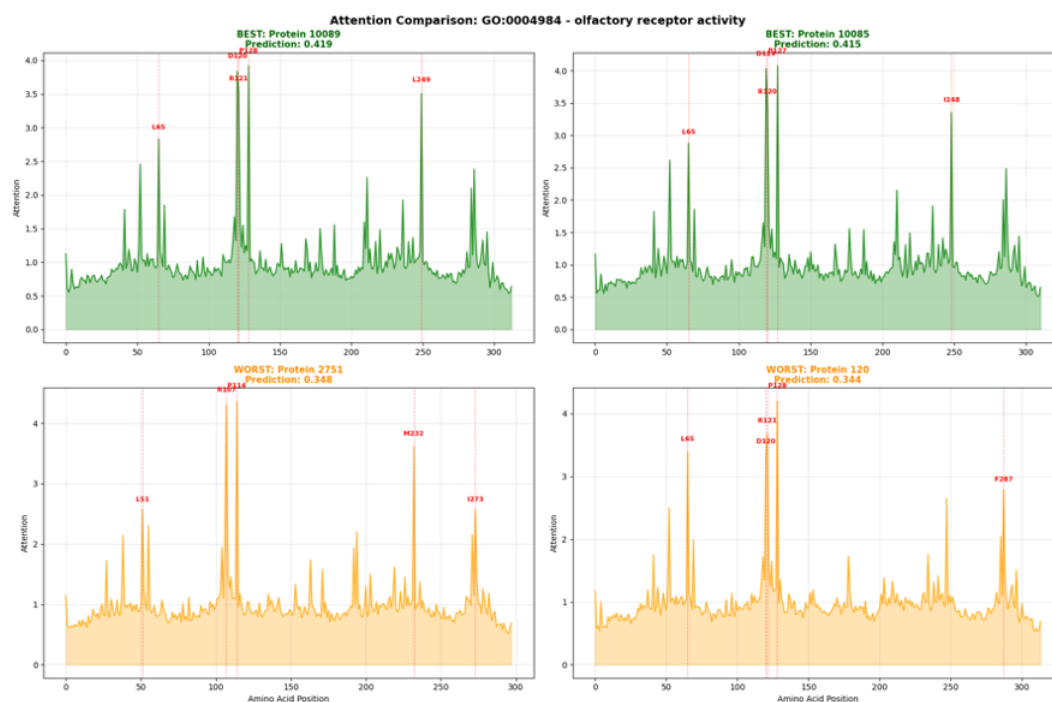


Figure 6: Attention profiles for olfactory receptor proteins. Top panels show attention distributions for two best-predicted proteins, characterized by sharp peaks at specific residue positions. Bottom panels show attention profiles for two worst-predicted proteins, exhibiting diffuse and noisy attention patterns.

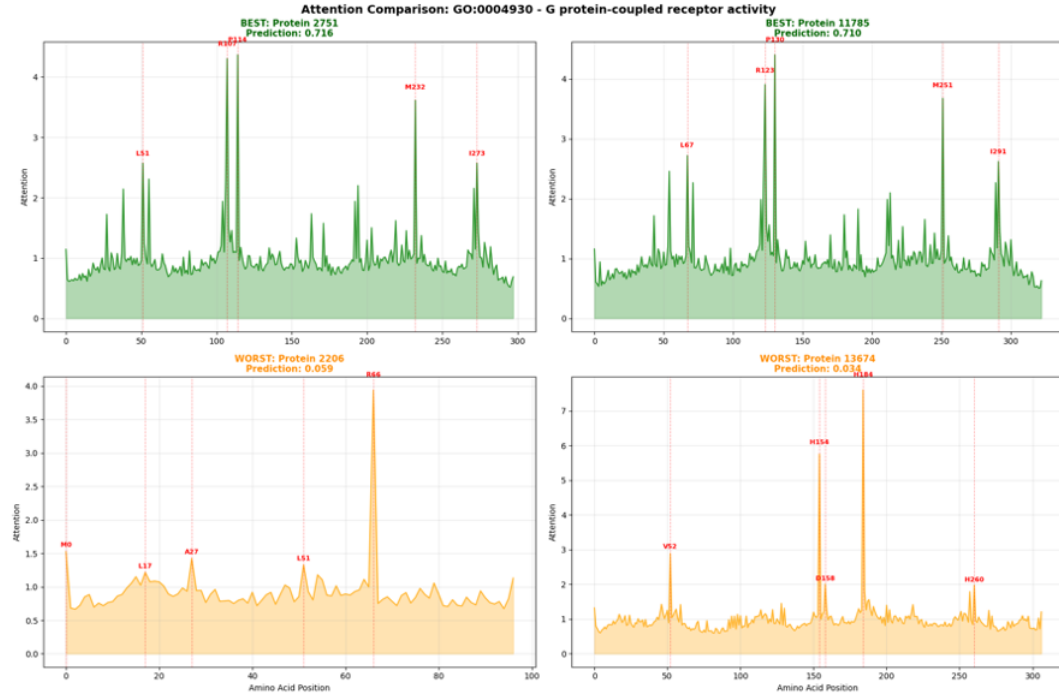


Figure 7: Attention profiles for G protein-coupled receptor proteins. Best-predicted proteins exhibit concentrated attention near conserved regions of the seven-transmembrane architecture, whereas worst-predicted proteins display dispersed attention without clear focal regions.

3.6.2 Integrated Gradients Analysis

Integrated gradients attribution revealed structured and function-specific usage of embedding dimensions within the 640-dimensional ESM-2 representation space. Across all 40 analyzed proteins, one embedding dimension (dimension 553) consistently appeared among the top five attributions, corresponding to a 100% occurrence rate. This dimension exhibited the highest mean absolute attribution value (0.42), indicating a strong and consistent contribution to model predictions across diverse protein families.

In addition to this shared component, distinct sets of embedding dimensions were associated with specific molecular function categories. Receptor proteins, including olfactory receptors (GO:0004984) and G protein-coupled receptors (GO:0004930), predominantly relied on dimensions 532, 565, and 493. Dimension 532 was present in all receptor proteins analyzed, while dimension 565 appeared in 88% of receptor examples. Kinase proteins (GO:0004672, GO:0004674, GO:0106310) showed a different signature, consistently activating dimensions 493, 187, and 568. Dimension 493 was observed in all kinase proteins, while dimension 187 appeared in 83% of cases.

DNA-binding transcription factors (GO:0000981) exhibited a separate dimensional profile, with primary contributions from dimensions 279, 350, and 385. Dimension 350 was present in 88% of transcription factor proteins and did not overlap with the dominant dimensions observed for receptors or kinases. Additional molecular function categories also displayed unique attribution patterns, including antigen-binding proteins (dimensions 291, 41, 529) and peptidases (dimensions 377, 515, 453).

Attribution patterns differed markedly between well-predicted and poorly predicted proteins. High-confidence predictions were characterized by consistent activation of function-specific embedding dimensions, whereas low-confidence predictions showed weaker or mixed activation patterns, often dominated by generic dimensions shared across multiple protein families.

Integrated gradients convergence diagnostics indicated reliable attribution estimates, with a mean convergence delta of 0.0012 across all proteins (range: -0.007 to $+0.003$), confirming stable gradient integration.

3.6.3 In Silico Mutagenesis

Systematic in silico mutagenesis was performed to assess whether high-attention residues identified by ESM-2 were functionally critical for model predictions. Three mutagenesis strategies were evaluated across three representative molecular function categories: olfactory receptors, G protein-coupled receptors, and protein kinases.

Single-residue alanine scanning produced minimal changes in predicted probabilities across all analyzed proteins, with prediction differences remaining below 5%. For olfactory receptor proteins, mutating the top five attention-ranked residues resulted in a maximum change of +0.5%. Similarly small effects were observed for GPCRs (-0.4%) and protein kinases (+4.3%).

Combinatorial mutagenesis involving simultaneous mutation of two to five residues also resulted in limited prediction changes across all protein families. Even when five residues were mutated simultaneously, prediction differences remained below 5% for olfactory receptors (-0.4%), GPCRs (-0.6%), and protein kinases (+3.7%).

In contrast, deletion-based mutagenesis revealed family-specific sensitivity patterns. For olfactory receptors, deletion of a five-residue window spanning positions 118–126 (centered on positions 120–121) resulted in a moderate prediction decrease of 5.5%, with similar effects observed across different window sizes. GPCR proteins exhibited high robustness, with the largest observed effect being a 2.2% decrease following deletion of a ten-residue window.

Protein kinases showed markedly higher sensitivity to deletions. Deletion of a five-residue window centered on position 220 resulted in a 10.7% reduction in predicted probability, while deletion of a ten-residue window centered on position 285 produced a 10.0% decrease. These were the only regions across all analyzed protein families where deletion resulted in prediction changes exceeding 10%.

A summary of deletion effects across window sizes and protein families is shown in Figure 8.

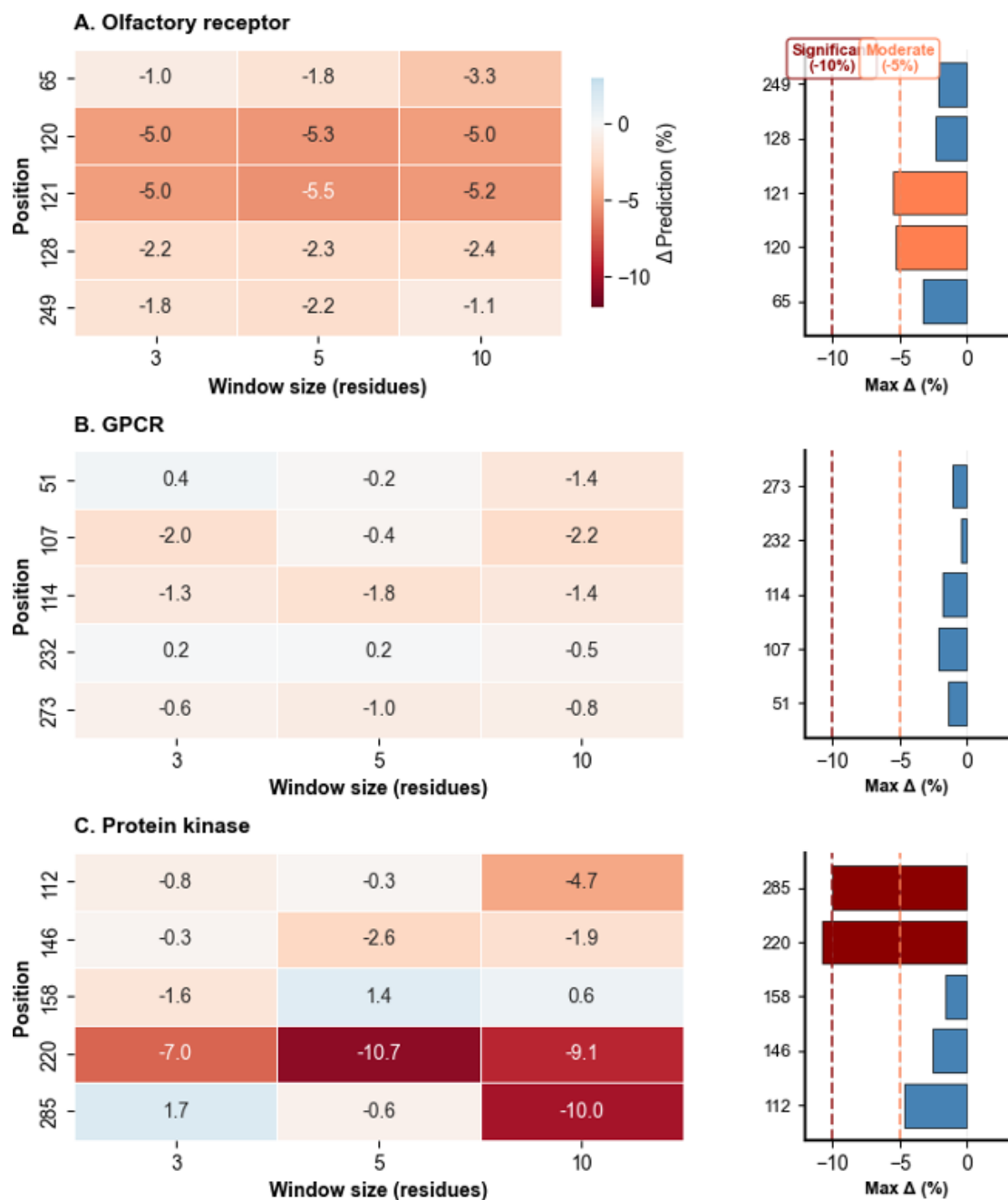


Figure 8: Deletion-based in silico mutagenesis reveals family-specific sensitivity patterns. Heatmaps (left) show prediction change (%) following deletion of 3-, 5-, or 10-residue windows centered on high-attention positions. Bar plots (right) report the maximum prediction change per position across all window sizes. Olfactory receptors exhibit moderate sensitivity to deletions, GPCRs remain highly robust, and protein kinases display localized regions with substantial prediction decreases exceeding 10%.

4 Discussion

4.1 Model Learning and Biological Validity

This work demonstrates that ESM-2 embeddings enable accurate protein molecular function prediction while maintaining biological interpretability. The $8.4\text{--}23\times$ improvement over random baselines indicates that the model learned genuine sequence–function relationships rather than exploiting dataset artifacts or label frequency statistics.

4.2 Interpretability of ESM-2 Representations

Integrated gradients analysis revealed that distinct protein families consistently activate specific dimensions within the 640-dimensional ESM-2 embedding space. Protein kinases preferentially relied on dimension 493, receptors activated dimensions 532 and 565, and DNA-binding transcription factors utilized dimensions such as 350 and 385. These dimension-specific signatures emerged without explicit supervision, suggesting that ESM-2 pre-training naturally organizes proteins according to underlying biochemical and structural constraints.

Importantly, these function-specific dimensions were used consistently by well-predicted proteins, whereas poorly predicted proteins showed mixed or nonspecific dimensional usage. This indicates that prediction confidence reflects the clarity of functional signatures encoded in the embedding space.

4.3 Robustness to Sequence Perturbations

Initial mutagenesis experiments showed that single-residue and combinatorial mutations produced minimal changes in model predictions. While this could be interpreted as model insensitivity, such behavior is consistent with biological reality: natural proteins are evolutionarily robust to point mutations, with only a small fraction of variants resulting in functional disruption.

Deletion-based mutagenesis provided a clearer picture of functional sensitivity. Protein kinases exhibited significant prediction decreases when deletions targeted specific catalytic regions, including the glycine-rich ATP-binding loop and the activation loop. In contrast, receptors remained robust even to larger deletions, reflecting their reliance on global fold architecture rather than localized catalytic machinery. This differential fragility mirrors well-established biological differences between catalytic enzymes and binding proteins.

4.4 Attention as a Computational Strategy

Attention analysis revealed that high-attention residues do not necessarily correspond to mutationally sensitive positions. Instead, attention appears to highlight computational landmarks such as conserved regions, domain boundaries, or structural transitions that help orient the model within sequence space during embedding generation.

Integrated gradients provided a complementary view by identifying which embedding dimensions, rather than which sequence positions, drove classifier predictions. The link between attention and integrated gradients was validated through deletion scanning: removal of high-attention regions disrupted the function-specific embedding dimensions identified by integrated gradients, resulting in reduced prediction confidence.

4.5 Comparison with Existing Methods

While existing approaches such as DeepGO and DeepFRI achieve comparable predictive performance, they provide limited insight into the internal mechanisms driving their predictions. In contrast, the combination of attention visualization, integrated gradients, and in silico mutagenesis used here offers complementary perspectives on sequence-level focus, feature-level importance, and causal relevance. Together, these analyses demonstrate that ESM-2 captures fundamental principles of protein evolution and function rather than memorizing training patterns.

4.6 Limitations

Class imbalance remains a primary limitation. The strong correlation between training set size and PR-AUC ($\rho = 0.76$) indicates that annotation scarcity, rather than model capacity, is the dominant bottleneck for rare function prediction. Additionally, restricting the dataset to human proteins limits generalization claims, and excluding sequences longer than 500 amino acids biases the analysis toward smaller, single-domain proteins.

The interpretability analysis focused on a limited set of representative proteins, enabling detailed investigation but precluding statistical coverage of all functional categories. Future work should extend these analyses proteome-wide.

4.7 Implications for Protein Engineering and Variant Interpretation

These findings have direct implications for protein design and variant analysis. Attention alone is insufficient for identifying functionally critical residues; instead, attention should be combined with embedding-level attributions and evolutionary conservation. The identification of function-specific embedding dimensions suggests that pathogenic variants may disrupt specific latent features rather than individual residues. For example, variants affecting the kinase-associated embedding dimension may impair function even if they do not directly target canonical catalytic motifs.

5 Conclusions

This work demonstrates that protein language models enable accurate and interpretable molecular function prediction directly from amino acid sequence. Using frozen ESM-2 embeddings, the proposed framework achieves an 8–23× improvement over random baselines on Gene Ontology molecular function annotation, with particularly strong performance for protein families characterized by conserved sequence motifs, such as G protein-coupled receptors (PR-AUC 0.996) and kinases (PR-AUC 0.892). Beyond predictive accuracy, comprehensive interpretability analyses reveal that the model learns biologically meaningful representations. Distinct protein families consistently activate specific, selective directions within the ESM-2 embedding space, indicating emergent functional specialization without explicit supervision. Mutagenesis experiments further show that the model correctly captures biological robustness: most proteins tolerate single and combinatorial mutations, while catalytic proteins such as kinases exhibit sensitivity to deletions in essential functional loops.

Importantly, attention was found to highlight computational landmarks rather than functionally critical residues, and was therefore not a faithful explanation on its own. Integrated gradients provided complementary insight by identifying the embedding-level dimensions driving predictions, and deletion-based perturbations established a causal link between sequence regions, those dimensions, and functional predictions. Together, these results demonstrate that model robustness to mutation is a feature rather than a limitation, reflecting fundamental principles of protein biology in which function arises from global architecture rather than fragile dependence on individual residues.

A key limitation of the current approach is the strong dependence of predictive performance on annotation availability. The observed correlation between training set size and PR-AUC ($\rho = 0.76$) indicates that the annotation gap remains the primary bottleneck for rare function prediction. Nevertheless, this work establishes a general and interpretable framework for protein function annotation that emphasizes biological realism alongside predictive performance.

6 Future Directions

Several avenues could further extend this work. Addressing class imbalance remains a central challenge, particularly for rare molecular functions. Few-shot and meta-learning approaches may enable transfer of knowledge from well-represented protein families to under-annotated functions. Incorporating hierarchical loss functions that reflect Gene Ontology parent–child relationships could further improve learning efficiency and biological consistency.

The scope of the model could also be expanded through multi-task learning across Molecular Function, Biological Process, and Cellular Component ontologies, as well as through cross-species evaluation to assess generalization beyond the human proteome. Integrating sequence-based embeddings with structure-based features derived from AlphaFold predictions represents another promising direction for combining global architectural information with local structural detail. Scaling the approach to larger protein language models, such as higher-capacity ESM-2 variants or emerging models like ESM-3, may further enhance representational richness.

Future work on interpretability could move from raw embedding dimensions toward genuinely monosemantic features by training sparse autoencoders on the model's activations, addressing the superposition that entangles individual dimensions. Causal intervention experiments within the activation space, attention-flow analysis across transformer layers, and the reconstruction of circuits linking features across layers would provide deeper mechanistic insight into how the model computes function, moving beyond identifying what it represents.

Finally, this framework has direct applications in variant effect prediction, protein engineering, and drug discovery. Embedding-level perturbations could be used to identify pathogenic variants, guide rational protein design, and prioritize druggable regions in disease-relevant targets. More broadly, the integration of protein language models with principled interpretability techniques offers a path toward trustworthy AI systems that not only predict protein function but also provide biologically interpretable reasoning.

References

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